COMPRESSED ODD DYNAMICS AND A FINITE STRUCTURAL SOLUTION FRAMEWORK FOR THE COLLATZ PROBLEM

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Abstract. We present a structural solution framework for the Collatz problem via the compressed odd map

$$U(n) = \frac{3n+1}{2^{v_2(3n+1)}}.$$

By encoding trajectories as valuation templates σ, we derive an exact reconstruction identity and a cycle equation that convert the problem into a finite arithmetic classification over template space. A bounded structural sieve—defined by a near-critical constraint around $m \log_2 3$, a forward-growth constraint, and a non-flatness condition—reduces the template universe from 132,303 to 5,574 and then to 611. Exact arithmetic compatibility reduces the core further to five survivors. Each residual candidate is then explicitly verified to converge to 1, yielding the reduction chain

$$132{,}303 \to 5{,}574 \to 611 \to 5 \to 0.$$

The governing structure is thereby reduced to a finite, explicit, and auditable endpoint.

1. Introduction

The Collatz map $T : \mathbb{N} \to \mathbb{N}$ is defined by

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod 2, \\ 3n+1, & n \equiv 1 \pmod 2. \end{cases}$$

The central conjecture asserts that every positive integer eventually reaches the cycle

$$1 \to 2 \to 4 \to 1.$$

Instead of analyzing T directly, we study the compressed odd dynamics

$$U(n) = \frac{3n+1}{2^{v_2(3n+1)}},$$

where n is odd and $v_2(3n+1)$ is the 2-adic valuation of $3n+1$. This compression removes the trivial even steps and yields a deterministic dynamical system on the odd integers. The core strategy is to replace trajectories by valuation templates, reduce the search to a finite family of templates, eliminate those templates arithmetically, and then verify the remaining residual cases explicitly.

2. Valuation Templates

Let $(n_i)_{i \geq 0}$ denote the odd states of a compressed trajectory, and define

$$\sigma_i = v_2(3n_i + 1).$$

For a template of length m, we write

$$\sigma = (\sigma_0, \sigma_1, \ldots, \sigma_{m-1}).$$

Associated to σ are the total dyadic reduction

$$B = B(\sigma) := \sum_{i=0}^{m-1} \sigma_i$$

and the partial sums

$$S_0 := 0, \qquad S_r := \sum_{j=0}^{r-1} \sigma_j \quad (1 \le r \le m).$$

These quantities organize the exact reconstruction formulas derived in the next section.

3. Exact Reconstruction Identity

Proposition 3.1 (Exact reconstruction)**.** *For any starting odd integer n_0 and valuation template σ of length m, the mth odd state is given by*

$$n_m = \frac{3^m n_0 + \sum_{r=0}^{m-1} 3^{m-1-r} 2^{S_r}}{2^B}.$$

Proof. Each compressed step satisfies

$$3n_i + 1 = 2^{\sigma_i} n_{i+1}.$$

Thus

$$n_{i+1} = \frac{3n_i + 1}{2^{\sigma_i}}.$$

Iterating once gives

$$n_1 = \frac{3n_0 + 1}{2^{\sigma_0}},$$

and substituting recursively yields a linear expression in n_0 whose coefficient on n_0 is $3^m/2^B$. The additive terms accumulate with weights determined by the partial sums S_r, producing the stated identity. □

4. Cycle Equation

For later use, define the template polynomial

$$P(\sigma) := \sum_{r=0}^{m-1} 3^{m-1-r} 2^{S_r}$$

and the determinant

$$D(\sigma) := 2^B - 3^m.$$

Corollary 4.1 (Cycle constraint)**.** *If a trajectory is periodic with $n_m = n_0$, then*

$$n_0 = \frac{P(\sigma)}{D(\sigma)}.$$

In particular, any periodic template must satisfy $D(\sigma) \neq 0$ and $D(\sigma) \mid P(\sigma)$.

Proof. Set $n_m = n_0$ in Proposition 3.1 and solve for n_0. □

5. Finite Structural Reduction

5.1. **Full template universe.** We restrict the search to templates of length $3 \leq m \leq 6$ with entries $\sigma_i \in \{0, \ldots, 6\}$. This produces a bounded universe of 132,303 templates.

Table 1. Distribution of the full template universe

Length (m)	Template Count
3	342
4	2400
5	16 750
6	112 811
Total	132 303

5.2. **Near-critical filter.** A recurrent candidate must remain close to the balance line $m \log_2 3$. Accordingly, we impose

$$|B - m \log_2 3| \leq 1.0.$$

This near-critical filter reduces the universe to 5,574 templates.

Table 2. Near-critical counts ($\varepsilon = 1.0$)

Length (m)	Near-Critical Count
3	36
4	200
5	795
6	4543
Total	5574

5.3. **Structural core selector.** The refined structural core is defined by the following three conditions:

(1) **Near-critical balance:** $|B - m \log_2 3| \leq 0.72$.
(2) **Forward-growth constraint:** $\sigma_{i+1} \leq \sigma_i + 1$.
(3) **Non-flatness:** $\max(\sigma) \neq \min(\sigma)$.

5.4. **Structural core distribution.** Under these constraints, the structural core Σ_{core} contains 611 templates.

5.5. **Reduction summary.** The bounded template universe contracts as follows:

$$132{,}303 \to 5{,}574 \to 611.$$

Table 3. Structural core distribution

Length (m)	Core Count
3	9
4	54
5	79
6	469
Total	611

6. Arithmetic Admissibility

Definition 6.1 (Arithmetic compatibility)**.** Let σ be a template of length m with associated quantities B, $P(\sigma)$, and $D(\sigma) = 2^B - 3^m$ as in Section 4. We say that σ is *arithmetically admissible* if all of the following hold:

- $D(\sigma) \neq 0$,
- $D(\sigma) \mid P(\sigma)$,
- $n_0 = P(\sigma)/D(\sigma) > 0$,
- n_0 is odd.

7. Arithmetic Reduction

The 611 templates in Σ_{core} were evaluated against Definition 6.1. The resulting failure modes are summarized below.

Table 4. Failure-mode counts in the structural core

Failure Mode	Count
Divisibility failure	596
Non-positive n_0	4
Even n_0	6
Survivors	5
Total	611

Thus the structural core contracts further according to

$$611 \to 5.$$

8. Residual Templates and Values

The source text provided for conversion identified five residual cases but did not include the explicit template tuples in the corresponding display. What is fully specified in the source is the resulting set of starting values,

$$n_0 \in \{17, 37, 43, 107, 217\}.$$

These are the only values remaining after the arithmetic reduction in Section 7.

9. Full Residual Verification

Theorem 9.1. *Each residual starting value in Section 8 converges to* 1 *under the compressed odd map* U.

Proof by explicit trajectories. The surviving values are verified as follows:

- **17:** $17 \to 13 \to 5 \to 1$.
- **37:** $37 \to 7 \to 11 \to 17 \to 13 \to 5 \to 1$.
- **43:** $43 \to 65 \to 49 \to 37 \to 7 \to 11 \to 17 \to 13 \to 5 \to 1$.
- **217:** $217 \to 163 \to 245 \to 23 \to 35 \to 53 \to 5 \to 1$.
- **107:**

$$\begin{aligned} 107 \to 161 \to 121 \to 91 \to 137 \to 103 \to 155 \to 233 \to 175 \to 263 \to \\ 395 \to 593 \to 445 \to 167 \to 251 \to 377 \to 283 \to 425 \to 319 \to 479 \to \\ 719 \to 1079 \to 1619 \to 2429 \to 911 \to 1367 \to 2051 \to 3077 \to 577 \to \\ 433 \to 325 \to 61 \to 23 \to 35 \to 53 \to 5 \to 1. \end{aligned}$$

Therefore $\Sigma_{\text{arith}} = \varnothing$, and the residual reduction is

$$5 \to 0.$$

□

10. Main Theorem

Theorem 10.1. *No nontrivial trajectory survives within the reconstructed structural core under the reduction chain*

$$132{,}303 \to 5{,}574 \to 611 \to 5 \to 0.$$

Proof. By Sections 7 and 9, every template in the structural core is eliminated either by arithmetic incompatibility or by explicit convergence of the corresponding residual starting value. This exhausts the bounded search defined in Section 5. □

Appendix A. Structural Selector Sweep

The structural core is defined over $3 \le m \le 6$ and $\sigma_i \in \{0, \dots, 6\}$ subject to the following constraints:

- $|B - m \log_2 3| \le 0.72$,
- $\sigma_{i+1} \le \sigma_i + 1$,
- $\max(\sigma) \ne \min(\sigma)$.

The table below records the stability of the selector under varying tolerance ε.

Table 5. Selector sweep table

Tolerance (ε)	Near-Critical Count	Structural Core Count
1.0	5,574	684
0.8	5,249	618
0.6	5,118	582
0.5	3,242	366

References

[1] J. C. Lagarias, "The $3x + 1$ problem and its generalizations," *The American Mathematical Monthly*, vol. 92, no. 1, pp. 3–23, 1985.

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[3] T. Tao, "Almost all orbits of the Collatz map attain almost bounded values," *Forum of Mathematics, Pi*, vol. 10, e12, 2022.

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